Create an Inbound Rule to Allow Remote Access from any IP Address SOP

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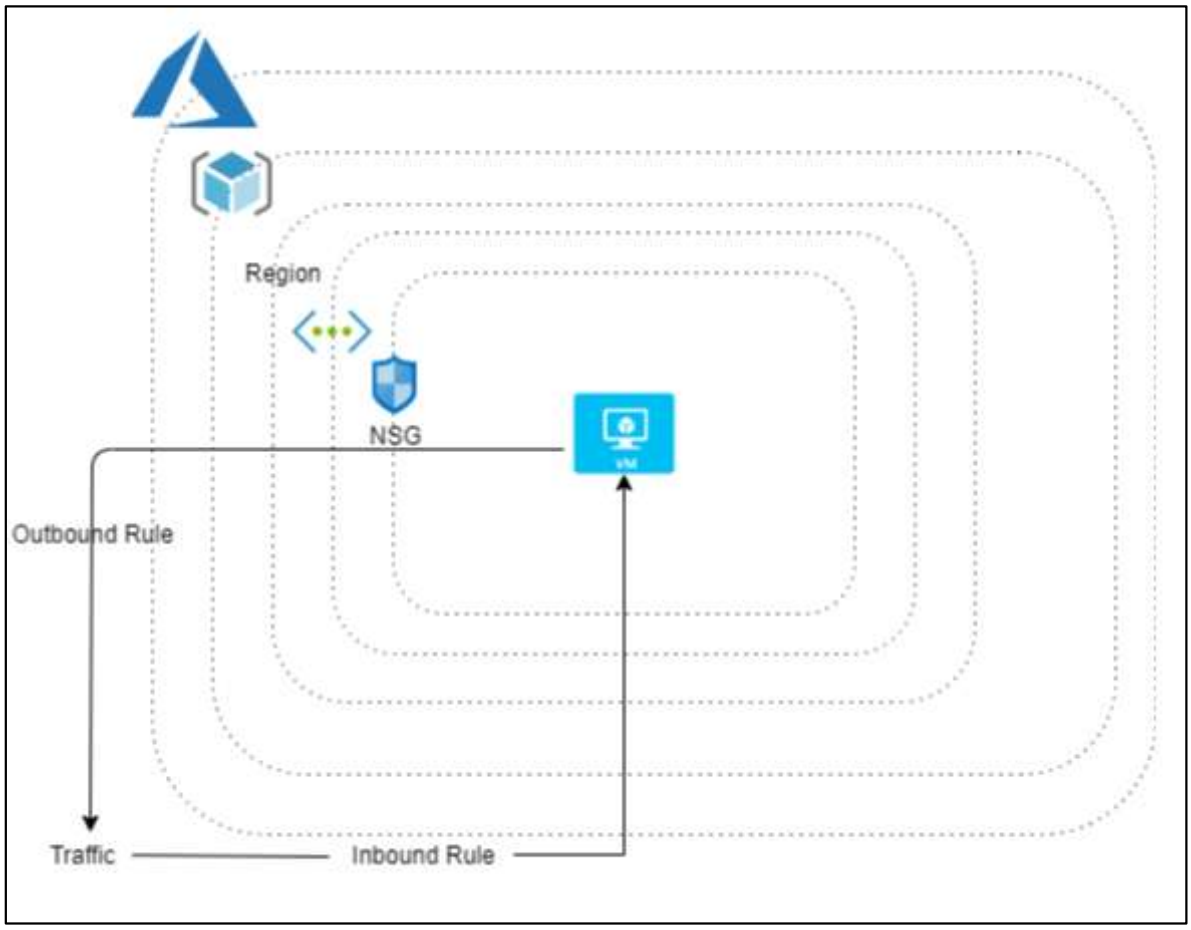
1. Create an Inbound Rule to Allow Remote Access from any IP Address

1.1 Description

Azure network security group filters network traffic to and from Azure resources in an Azure virtual network. A network security group contains security rules that allow or deny inbound network traffic to, or outbound network traffic from, several types of Azure resources. For each rule, you can specify source and destination, port, and protocol. To give remote desktop access to the virtual machine, you need to configure Inbound Port Rules in the network security group of the Virtual Machine.

1.2 Architecture Diagram

The diagram below displays a visual representation of the application architecture:



1.3 Lab Steps

Follow the steps outlined below to achieve the objective of this lab exercise:

1. Create a Virtual machine:
    a. Go to the Azure Portal and select **Create a resource** > **Compute** > **Virtual Machine**.
    b. Give the following details, and note that the values used here are sample ones.
        i. Give Resource Group: Your Resource Group name
        ii. Virtual Machine Name: labvm1
        iii. Region: Any region (Go as per the Lab policies)
        iv. Availability options: Leave it to its default value.
        v. Image: Windows Server 2016 Datacenter or Ubuntu Server 18.04 LTS
        vi. Size: Standard_B1s
        vii. Provide the username and password.
        viii. Public Inbound ports: None
        ix. Leave other fields with their default values.
        x. In the **Management blade**, select **Boot diagnostics**: **Off**
        xi. Leave other fields with their default values, and click **Review+Create** > **Create**.

2. Connect to Virtual Machine:
    a. Go to the Virtual Machine created by you.
    b. In the **overview** section, click on **Connect** and select **RDP for Windows VM**.
    c. Download the RDP file and try to connect with your username and password.
    Note: You will get an error, since the Traffic on RDP port is not allowed.
    Similarly, for Linux VM, if you try to access the machine via SSH, you will get an error since traffic not allowed on port 22.

3. Create an Inbound Rule:
    a. Go to the virtual machine created by you.
    b. Go to the **Networking** Section under **Settings**.
    c. In the **Inbound Port rules**, click on **Add Inbound Port rule**.
    d. Provide the following details:
        i. Source: Any
        ii. Source Port Ranges: Any
        iii. Destination: Any
        iv. Destination Port Ranges: 3389 (for Windows VM), 22 (for Linux VM)
        v. Protocol: Any
        vi. Action: Allow
        vii. Priority: 110
        viii. Name: Port_3389 (for Windows VM), Port_22 (for Linux VM)
    e. Click on **Add**.
    f. Connect to the instance and remain connected.
    g. Again go to the **Networking** Section and add another inbound rule, provide the following details:
        i. Source: Any
        ii. Source Port Ranges: Any
        iii. Destination: Any
        iv. Destination Port Ranges: *
        v. Protocol: Any
        vi. Action: Deny

vii. Priority: 100
viii. Name: Port_Any

h. If you try to make a new connection to the VM, it will not be allowed.

4. Clean up Resources:
   a. When you complete the lab exercise, delete all the resources created by you.

1.4 Troubleshooting

S. No	Problem	Solution
1	Connection still Intact	Your existing connection may still remain intact, but a new connection will not be allowed after creating Deny Traffic Rule.
2	Connection not established	Allowing or Denying of Connection depends on Priority, check the priority of Rules.
3	Location disallowed by Policy	Go as per the Lab Policy, look for any error and suggested options to be followed.

1.5 Supporting References

Refer the below links for additional information:

1. https://docs.microsoft.com/bs-latn-ba/azure/virtual-machines/troubleshooting/troubleshoot-rdp-nsg-problem
2. https://docs.microsoft.com/en-us/azure/virtual-machines/windows/quick-create-portal